Logarithmic algebraic geometry: fine/fs log schemes and their fiber products

Simon Felten

November 4, 2026

1 Charts and coherence

Definition 1.1. Let $(X, \mathcal{M}_X)$ be a log scheme.

(1) Let $W \subseteq X$ be an open subset. A *chart* on W *subordinate* to a monoid P is a monoid homomorphism $\beta\colon P \to \Gamma(W, \mathcal{M}_X)$ such that the middle vertical map δ in the diagram

$$\begin{array}{ccccc} \underline{P}_W & \longrightarrow & (\underline{P}_W)^{\log} & \longrightarrow & \mathcal{O}_W \\ \| & & \downarrow{\scriptstyle \delta} & & \| \\ \underline{P}_W & \xrightarrow{\beta} & \mathcal{M}_W & \xrightarrow{\alpha_X} & \mathcal{O}_W \end{array}$$

is an isomorphism. Here, δ is induced by the universal property of logification.

(2) The log structure $\mathcal{M}_X$ is *quasi-coherent* if it locally admits a chart. We write $\mathrm{Sch}^{\log}_{\mathrm{qc}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are quasi-coherent log schemes.

(3) The log structure $\mathcal{M}_X$ is *coherent* if it locally admits a chart subordinate to a finitely generated monoid. We write $\mathrm{Sch}^{\log}_{\mathrm{coh}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are coherent log schemes.

(4) The log structure $\mathcal{M}_X$ is *fine* if it coherent and integral. We write $\mathrm{Sch}^{\log}_{\mathrm{fine}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are fine log schemes.

(5) The log structure $\mathcal{M}_X$ is *fine and saturated* (*fs*) if it coherent, integral, and saturated. We write $\mathrm{Sch}^{\log}_{\mathrm{fs}}$ for the full subcategory of $\mathrm{Sch}^{\log}$ whose objects are fine and saturated log schemes. ⧫

Example 1.2. Let $P \to R$ be a pre-log ring. Then, $\mathrm{Spec}(P \to R)$ is quasi-coherent. If P is finitely generated, then $\mathrm{Spec}(P \to R)$ is coherent. ◊

Remark 1.3. Let $\mathcal{M}_X$ be a log structure. Then, it is *not* true that $\mathcal{M}_X$ is fine as a log structure if and only if its stalks $\mathcal{M}_{X,x}$ are fine as monoids. Indeed, these are very different notions. ◊

Fact 1.4. *Let $(X, \mathcal{M}_X)$ be a log scheme.*

(1) *Assume that $\mathcal{M}_X$ admits a chart subordinate to a fine (respectively fs) monoid P. Then, $\mathcal{M}_X$ is integral (integral and saturated) and thus fine (respectively fs).*

(2) *Assume that $\mathcal{M}_X$ is fine (respectively fs), and let $x \in X$. Then, we have a fine (respectively fs) monoid P, an open neighborhood $x \in W \subseteq X$, and a chart $\beta\colon P \to \Gamma(W, \mathcal{M}_X)$ such that the induced map $\overline{P} \to \overline{\mathcal{M}}_{X,x}$ is an isomorphism. We say that the chart is* exact *at x.*

Example 1.5. Let $(X, \mathcal{M}_X) = (A_{\mathbb{N}}|D_{\mathbb{N}})$. Then, the chart $\mathbb{N} \to \Gamma(X, \mathcal{M}_X)$ is exact at $x = 0$ but not exact at any other point. ◊

For a general commutative monoid P, we write $A_P^\dagger \coloneqq \mathrm{Spec}(P \to \mathbb{Z}[P])$.

Fact 1.6. *Let $f\colon X \to Y$ be a morphism of schemes, and let $\beta\colon \mathcal{P}_Y \to \mathcal{O}_Y$ be a pre-log structure. Let $\mathcal{M}_Y = (\mathcal{P}_Y)^{\log}$. Then, $(f^{-1}\mathcal{P}_Y)^{\log} \to (f^{-1}\mathcal{M}_Y)^{\log}$ is an isomorphism of log structures on X.*

Lemma 1.7. *Let $(X, \mathcal{M}_X)$ be a log scheme, and let $W \subseteq X$ be an open subset. Then, the datum of a chart $\beta\colon P \to \Gamma(W, \mathcal{M}_X)$ for $\mathcal{M}_X$ on W is equivalent to the datum of a strict morphism of log schemes $(W, \mathcal{M}_W) \to A_P^\dagger$.*

Proof. Let $\beta\colon P \to \Gamma(W, \mathcal{M}_X)$ be a monoid homomorphism. Then, we obtain a homomorphism of pre-log rings $(P \to \mathbb{Z}[P]) \to (\Gamma(W, \mathcal{M}_X) \to \Gamma(W, \mathcal{O}_X))$ and hence a morphism of log schemes $\phi\colon (W, \mathcal{M}_W) \to A_P^\dagger$ by adjunction. Conversely, if $\phi\colon (W, \mathcal{M}_W) \to A_P^\dagger$ is a morphism, then $\phi^\flat$ induces a map $\beta\colon P \to \Gamma(A_P, \mathcal{M}_{A_P}) \to \Gamma(W, \mathcal{M}_X)$. These two constructions are inverse to each other.

It remains to show that ϕ is strict if and only if β is a chart. Now, $(\underline{P}_W)^{\log} \to \phi_{\log}^* \mathcal{M}_{A_P}$ is an isomorphism, so the claim follows. □

Fact 1.8 (Charts for morphisms of coherent log schemes)**.** *Let $f\colon (X, \mathcal{M}_X) \to (Y, \mathcal{M}_Y)$ be a morphism of coherent log schemes, and let $x \in X$. For every open neighborhood $f(x) \in Y^\circ \subseteq Y$ and every chart $\beta\colon Q \to \Gamma(Y^\circ, \mathcal{M}_Y)$ with Q finitely generated, there is an open neighborhood $x \in X^\circ \subseteq f^{-1}(Y^\circ)$ and a commutative diagram*

$$\begin{array}{ccc} P & \xrightarrow{\gamma} & \Gamma(X^\circ, \mathcal{M}_X) \\ {\scriptstyle\theta}\uparrow & & \uparrow{\scriptstyle f^\flat} \\ Q & \xrightarrow{\beta} & \Gamma(Y^\circ, \mathcal{M}_Y) \end{array}$$

with a finitely generated monoid P and such that $\gamma\colon P \to \Gamma(X^\circ, \mathcal{M}_X)$ is a chart for $\mathcal{M}_{X^\circ}$.

Remark 1.9. When $\mathcal{M}_X$ is not coherent but only quasi-coherent, the conclusion remains true except that we may not choose P finitely generated. If Q is not finitely generated, it is (to the author's knowledge) not known if the conclusion holds. For a discussion of these issues, see [1]. ◊

Fact 1.10 (Exact charts for morphisms of fine/fs log schemes)**.** *Let $f\colon (X, \mathcal{M}_X) \to (Y, \mathcal{M}_Y)$ be a morphism of fine (fs) log schemes, and let $x \in X$. For every open neighborhood $f(x) \in Y^\circ \subseteq Y$ and every chart $\beta\colon Q \to \Gamma(Y^\circ, \mathcal{M}_Y)$ with Q fine (fine and saturated), there is an open neighborhood $x \in X^\circ \subseteq f^{-1}(Y^\circ)$ and a commutative diagram*

$$\begin{array}{ccc} P & \xrightarrow{\gamma} & \Gamma(X^\circ, \mathcal{M}_X) \\ {\scriptstyle\theta}\uparrow & & \uparrow{\scriptstyle f^\flat} \\ Q & \xrightarrow{\beta} & \Gamma(Y^\circ, \mathcal{M}_Y) \end{array}$$

with a fine (fine and saturated) monoid P and such that $\gamma\colon P \to \Gamma(X^\circ, \mathcal{M}_X)$ is a chart for $\mathcal{M}_{X^\circ}$ which is exact at x, i.e., $\overline{P} \to \overline{\mathcal{M}}_{X,x}$ is an isomorphism.

2 Fine and fs fiber products

Let us first show that the fiber product of coherent log schemes is coherent.

Lemma 2.1. *Let $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ and $b\colon (T, \mathcal{M}_T) \to (S, \mathcal{M}_S)$ be two morphisms of coherent log schemes. Consider the fiber product*

$$\begin{array}{ccc} (Y, \mathcal{M}_Y) & \xrightarrow{c} & (X, \mathcal{M}_X) \\ \downarrow{\scriptstyle g} & \overset{r}{\searrow} & \downarrow{\scriptstyle f} \\ (T, \mathcal{M}_T) & \xrightarrow{b} & (S, \mathcal{M}_S) \end{array}$$

in $\mathrm{Sch}^{\log}$. *Then, $(Y, \mathcal{M}_Y)$ is coherent.*

Proof. After shrinking X, S, and T, we may assume that we have charts $Q \to \Gamma(S, \mathcal{M}_S)$, $P_1 \to \Gamma(X, \mathcal{M}_X)$, and $P_2 \to \Gamma(T, \mathcal{M}_T)$, as well as compatible monoid homomorphisms $\theta_1\colon Q \to P_1$ and $\theta_2\colon Q \to P_2$. Then, we have charts $\underline{Q} \to r^*_{\log}\mathcal{M}_S$, $\underline{P}_1 \to c^*_{\log}\mathcal{M}_X$, and $\underline{P}_2 \to g^*_{\log}\mathcal{M}_T$. Now, one may show that

$$\underline{P_1 \oplus_Q P_2} \to \mathcal{M}_Y$$

is a chart as well, by studying the universal property of logification.

More precisely, let Log_Y be the category of log structures on Y, and let PLog_Y be the category of pre-log structures on Y. Then, $(-)^{\log}\colon \mathrm{PLog}_Y \to \mathrm{Log}_Y$ is left adjoint to the inclusion $\mathrm{Log}_Y \to \mathrm{PLog}_Y$, so it transforms colimits in PLog_Y into colimits in Log_Y. □

However, fiber products of fine log schemes are not necessarily fine:

Example 2.2. This example is a variant of [2, Ex. 0.3]. Let $Q = Q^a = \mathbb{N}^{\oplus 2}$, $P_1 = P_1^a = \mathbb{N}\cdot(1,0) \oplus \mathbb{N}\cdot(-1,1)$, and $P_2 = P_2^a = \mathbb{N}\cdot(0,1) \oplus \mathbb{N}\cdot(1,-1)$. Let $\theta_1\colon Q \to P_1$ and $\theta_2\colon Q \to P_2$ be given by the inclusions. First, we compute

$$\begin{aligned} \mathbb{C}[P_1] \otimes_{\mathbb{C}[Q]} \mathbb{C}[P_2] &\cong \mathbb{C}[x,z] \otimes_{\mathbb{C}[x,y]} \mathbb{C}[y,w] \\ &\cong \mathbb{C}[x,y,z,w]/(xz-y, yw-x) \cong \mathbb{C}[y,z,w]/(yzw-y). \end{aligned}$$

Let $P = P_1 \oplus_Q P_2$, and let $\iota_P\colon P \to P^{\mathrm{gp}}$ be the canonical map. We have

$$P^{\mathrm{gp}} = P_1^{\mathrm{gp}} \oplus_{Q^{\mathrm{gp}}} P_2^{\mathrm{gp}} = \mathbb{Z}^2 \oplus_{\mathbb{Z}^2} \mathbb{Z}^2 = \mathbb{Z}^2.$$

Let $\bar{x} = (1,0) \in Q$, $\bar{y} = (0,1) \in Q$, $\bar{z} = (-1,1) \in P_1$, and $\bar{w} = (1,-1) \in P_2$. In P^{gp}, we have $\iota_P(\bar{z}) + \iota_P(\bar{w}) = 0$, so if ι_P is injective, we have $\bar{z} + \bar{w} = 0$ in P. In particular, we have $1 = \beta(\bar{z}) \cdot \beta(\bar{w}) = zw \in \mathbb{C}[P]$. However, this is not the case, as $\mathbb{C}[P] \cong \mathbb{C}[y,z,w]/(yzw-y)$. Thus, $\iota_P\colon P \to P^{\mathrm{gp}}$ is not injective, and P is not integral. The same argument also shows that $\mathcal{M}_Y$ is not integral, for $Y^\dagger = \mathrm{Spec}(P \to \mathbb{C}[P])$ the fiber product. Namely, we cannot have $\bar{z} + \bar{w} = 0$ in $\Gamma(Y, \mathcal{M}_Y)$. ◊

In this section, we will construct alternative fiber products $X^\dagger \times^{\mathrm{fine}}_{S^\dagger} T^\dagger$ and $X^\dagger \times^{\mathrm{fs}}_{S^\dagger} T^\dagger$ which are fine respectively fine and saturated.

The inclusion $\mathrm{CMon}^{\mathrm{int}} \to \mathrm{CMon}$ of integral monoids into the category of all commutative monoids admits a left adjoint given by

$$P \mapsto P^{\mathrm{int}} := \mathrm{im}(i_P \colon P \to P^{\mathrm{gp}}).$$

Example 2.3. (1) We resume the previous example $Q = Q^a$ etc. Since $P_1 \oplus_Q P_2$ is a quotient of $P_1 \oplus P_2$, we have

$$(P_1 \oplus_Q P_2)^{\mathrm{int}} = \mathrm{im}(P_1 \oplus P_2 \to \mathbb{Z}^{\oplus 2}) = \langle (1,0),(0,1),(1,-1),(-1,1)\rangle = \mathbb{N}\cdot(1,0) \oplus \mathbb{Z}\cdot(1,-1).$$

(2) Let $Q = Q^b = \mathbb{N}$, $P_1 = P_1^b = \mathbb{N}$, and $P_2 = P_2^b = \mathbb{N}$. Let $\theta_1(1) = 2$ and $\theta_2(1) = 2$. The pushout $P = P_1 \oplus_Q P_2$ is integral, but we do not know this yet. Nonetheless, we can compute P^{int} as before. We have $P^{\mathrm{gp}} = \mathbb{Z} \oplus_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}^{\oplus 2}/\mathbb{Z}\cdot(-2,2) \cong \mathbb{Z} \oplus \mathbb{Z}/2$. The last homomorphism is given by $(1,0) \mapsto (1,0)$ and $(-1,1) \mapsto (0,1)$. Inside P^{gp}, P^{int} is generated by $(1,0)$ and $(0,1)$, so in $\mathbb{Z} \oplus \mathbb{Z}/2$, it is generated by $(1,0)$ and $(1,1)$. $\lozenge$

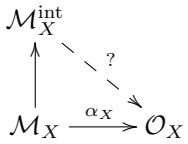
Similarly, the inclusion $\mathrm{CMon}^{\mathrm{sat}} \to \mathrm{CMon}$ of saturated monoids into the category of all commutative monoids admits a left adjoint given by

$$P \mapsto P^{\mathrm{sat}} = \{a \in P^{\mathrm{gp}} \mid \exists\, k \geq 1, p \in P : ka = i_P(p)\}.$$

Example 2.4. (1) Let $P = \langle (2,0),(0,3),(1,2)\rangle$. Then, we have $P^{\mathrm{sat}} = \mathbb{N}^{\oplus 2}$.

(2) Let $Q = Q^b$ etc. Then, P^{int} is not saturated because $2\cdot(0,1) = (0,0) \in P^{\mathrm{int}}$ but $(0,1) \notin P^{\mathrm{int}}$. We have $P^{\mathrm{sat}} \cong \mathbb{N} \oplus \mathbb{Z}/2$. $\lozenge$

We cannot simply apply $(-)^{\mathrm{int}}$ to $\mathcal{M}_X$ in order to obtain an integral log structure. Namely, $\mathcal{M}_X^{\mathrm{int}}$ does not necessarily admit a map to $\mathcal{O}_X$, so we may not obtain a log structure this way:



Thus, we also have to modify $\mathcal{O}_X$, and then we need to modify the underlying topological space.

Proposition 2.5. *The following statements hold:*

(1) *The inclusion* $\mathrm{Sch}^{\log}_{\mathrm{qc,int}} \to \mathrm{Sch}^{\log}_{\mathrm{qc}}$ *admits a right adjoint* $(X,\mathcal{M}_X) \mapsto (X,\mathcal{M}_X)^{\mathrm{int}}$. *If* $(X,\mathcal{M}_X)$ *is coherent, then* $(X,\mathcal{M}_X)^{\mathrm{int}}$ *is fine. For a commutative monoid* P, *we have* $(A_P^\dagger)^{\mathrm{int}} = A^\dagger_{P^{\mathrm{int}}}$. *The morphism*

$$(X,\mathcal{M}_X)^{\mathrm{int}} \to (X,\mathcal{M}_X)$$

is a closed immersion on underlying schemes.

(2) *The inclusion* $\mathrm{Sch}^{\log}_{\mathrm{qc,sat}} \to \mathrm{Sch}^{\log}_{\mathrm{qc}}$ *admits a right adjoint* $(X,\mathcal{M}_X) \mapsto (X,\mathcal{M}_X)^{\mathrm{sat}}$. *If* $(X,\mathcal{M}_X)$ *is coherent, then* $(X,\mathcal{M}_X)^{\mathrm{sat}}$ *is fs. For a commutative monoid* P, *we have* $(A_P^\dagger)^{\mathrm{sat}} = A^\dagger_{P^{\mathrm{sat}}}$. *For a coherent log scheme* $(X,\mathcal{M}_X)$, *the morphism*

$$(X,\mathcal{M}_X)^{\mathrm{sat}} \to (X,\mathcal{M}_X)$$

is finite on underlying schemes. If $(X,\mathcal{M}_X)$ *is fine, then* $(X,\mathcal{M}_X)^{\mathrm{sat}} \to (X,\mathcal{M}_X)$ *is surjective.*

Proof. We only treat the case $(-)^{\mathrm{int}}$, the other case being similar. First, we show that every morphism $Z^\dagger \to A_P^\dagger$ from a quasi-coherent m-integral log scheme $Z^\dagger$ factors uniquely through $A_{P^{\mathrm{int}}}^\dagger \to A_P^\dagger$. Indeed, $\Gamma(Z, \mathcal{M}_Z)$ is integral, so we have a unique factorization $P \to P^{\mathrm{int}} \to \Gamma(Z, \mathcal{M}_Z)$, and then $P^{\mathrm{int}} \to \Gamma(Z, \mathcal{M}_Z)$ gives rise to the map $Z^\dagger \to A_{P^{\mathrm{int}}}^\dagger$. Since $P \to P^{\mathrm{int}}$ is surjective, $A_{P^{\mathrm{int}}}^\dagger \to A_P^\dagger$ is a closed immersion. If P is finitely generated, then P^{int} is fine, and $A_{P^{\mathrm{int}}}^\dagger$ is a fine log scheme.

Next, let $W^\dagger \to A_P^\dagger$ be strict. Then, we form the Cartesian square

$$\begin{array}{ccc} (W^\dagger)^{\mathrm{int}} & \longrightarrow & W^\dagger \\ \downarrow & & \downarrow \\ A_{P^{\mathrm{int}}}^\dagger & \longrightarrow & A_P^\dagger. \end{array}$$

The universal property of the Cartesian square shows that, for every quasi-coherent m-integral log scheme $Z^\dagger$ and morphism $Z^\dagger \to W^\dagger$, there is a unique factorization through $(W^\dagger)^{\mathrm{int}} \to W^\dagger$.

Finally, let $X = \bigcup_i W_i$, and assume that we have strict morphisms $W_i^\dagger \to A_{P_i}^\dagger$. Then, we obtain morphisms $(W_i^\dagger)^{\mathrm{int}} \to W_i^\dagger$ with the expected universal property. In particular, we have a canonical isomorphism $(W_i^\dagger)^{\mathrm{int}}|_{W_i \cap W_j} \cong (W_j^\dagger)^{\mathrm{int}}|_{W_i \cap W_j}$, and hence they glue to a scheme $(X^\dagger)^{\mathrm{int}} \to X^\dagger$. The universal property is induced by the universal property of the $(W_i^\dagger)^{\mathrm{int}} \to W_i^\dagger$. □

Definition 2.6. We define the following two fiber products:

(1) Let $f\colon X^\dagger \to S^\dagger$ and $b\colon T^\dagger \to S^\dagger$ be two morphisms of fine log schemes. Then, we set

$$X^\dagger \times_{S^\dagger}^{\mathrm{fine}} T^\dagger := (X^\dagger \times_{S^\dagger} T^\dagger)^{\mathrm{int}},$$

a closed subset of the usual fiber product. This is a fine log scheme, and the resulting square is Cartesian in $\mathrm{Sch}_{\mathrm{fine}}^{\log}$.

(2) Let $f\colon X^\dagger \to S^\dagger$ and $b\colon T^\dagger \to S^\dagger$ be two morphisms of fs log schemes. Then, we set

$$X^\dagger \times_{S^\dagger}^{\mathrm{fs}} T^\dagger := (X^\dagger \times_{S^\dagger} T^\dagger)^{\mathrm{sat}}.$$

This is an fs log scheme, and the resulting square is Cartesian in $\mathrm{Sch}_{\mathrm{fs}}^{\log}$. The morphism to the usual fiber product is finite. ⧫

Remark 2.7. We do not know if the fiber product of quasi-coherent log schemes is quasi-coherent, so we do not make this definition in the quasi-coherent case. ◊

Example 2.8. Fine and fs fiber products can be empty. For an example, consider a commutative diagram (not necessarily Cartesian)

$$\begin{array}{ccc} Z^\dagger & \xrightarrow{c} & X^\dagger = \mathrm{Spec}(P_1 \to \mathbb{C}) \\ \downarrow{\scriptstyle g} & \overset{r}{\searrow} & \downarrow{\scriptstyle f} \\ T^\dagger = \mathrm{Spec}(P_2 \to \mathbb{C}) & \xrightarrow{b} & S^\dagger = \mathrm{Spec}(\mathbb{N}^{\oplus 2} \to \mathbb{C}) \end{array}$$

with

$$P_1 = P_1^a = \mathbb{N} \cdot (1,0) \oplus \mathbb{N} \cdot (-1,1) \quad \text{and} \quad P_2 = P_2^a = \mathbb{N} \cdot (0,1) \oplus \mathbb{N} \cdot (1,-1)$$

and $\mathcal{M}_Z$ integral. Let $\bar{w} = (1,-1) \in \mathcal{M}_X$, $\bar{z} = (-1,1) \in \mathcal{M}_T$, and $\bar{x} = (1,0), \bar{y} \in (0,1) \in \mathcal{M}_S$. Then, we have $g^\flat(\bar{z}) + r^\flat(\bar{x}) = r^\flat(\bar{y})$ and $c^\flat(\bar{w}) + r^\flat(\bar{y}) = r^\flat(\bar{x})$. Since $\mathcal{M}_Z$ is integral, we have $c^\flat(\bar{w}) + g^\flat(\bar{z}) = 0$. Thus, we have

$$1 = \alpha_Z(c^\flat(\bar{w})) \cdot \alpha_Z(g^\flat(\bar{z})) = (c^\sharp \circ \alpha_X(\bar{w})) \cdot (g^\sharp \circ \alpha_T(\bar{z})) = 0.$$

This is only possible if $Z = \varnothing$. In particular, the fine and fs fiber products are empty. ◊

3 Exact morphisms

The following notion is often used in log geometry. Currently, our main motivation is that it has a pleasant consequence for the fine fiber product.

Definition 3.1. A monoid homomorphism $\theta\colon Q \to P$ is *exact* if

$$\begin{array}{ccc} Q & \xrightarrow{\theta} & P \\ \downarrow & & \downarrow \\ Q^{\mathrm{gp}} & \longrightarrow & P^{\mathrm{gp}} \end{array}$$

is Cartesian. It is *locally exact* if, for every face $F \subseteq P$, the localized homomorphism $\theta\colon Q_{\theta^{-1}(F)} \to P_F$ is exact. ♦

Fact 3.2. *The following statements hold:*

(1) *Let $\theta\colon Q \to P$ and $\xi\colon P \to R$ be exact homomorphisms. Then, $\xi \circ \theta\colon Q \to R$ is exact.*

(2) *A homomorphism $\theta\colon Q \to P$ of integral monoids is exact if and only if $\overline{Q} \to \overline{P}$ is exact.*

(3) *Let $\theta\colon Q \to P$ be an injective homomorphism of integral monoids. Then, θ is exact if and only if $P \setminus Q$ is stable under $Q \times P \to P$.*

(4) *Let $\theta\colon Q \to P$ be an exact homomorphism of integral monoids, and let $\psi\colon Q \to Q'$ be another homomorphism of integral monoids. Then, $\theta'\colon Q' \to (Q' \oplus_Q P)^{\mathrm{int}}$ is exact.*

Example 3.3. (1) Let $Q = \mathbb{N}$, $P = \mathbb{N}$, and $\theta(1) = 3$. Then, $\mathbb{N} \setminus 3\mathbb{N}$ is stable under $3\mathbb{N} \times \mathbb{N} \to \mathbb{N}$, so θ is exact.

(2) Let $Q = \mathbb{N}^{\oplus 2}$ and $P = \mathbb{N} \cdot (1,0) \oplus \mathbb{N} \cdot (-1,1)$. Then, $\theta\colon Q \to P$ is not exact. Namely, we have $(-1,1) \notin Q$ but $(-1,1) + (1,0) \in Q$.

(3) The homomorphism $\theta\colon \mathbb{N} \to \mathbb{Z}$ is locally exact but not exact.

(4) The homomorphism $\theta\colon \mathbb{N}^{\oplus 2} \to \mathbb{N}^{\oplus 3}$ given by $\theta(a,b) = (a, a+b, b)$ is exact but not locally exact. Namely, $Q_{\theta^{-1}(F)} \to P_F$ is not exact for $F = \mathbb{N} \oplus 0 \oplus 0$. ◊

Definition 3.4. A morphism of fine log schemes $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ is *exact* if $f^\flat\colon \mathcal{M}_{S,f(x)} \to \mathcal{M}_{X,x}$ is exact for every $x \in X$. ♦

Example 3.5. Let $\theta\colon Q \to P$ be a homomorphism of toric monoids. Then, $A_P^\dagger \to A_Q^\dagger$ is exact if and only if $\theta\colon Q \to P$ is locally exact. $\lozenge$

For our current purposes, the following result is interesting.

Fact 3.6. *Let $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ be an exact morphism of fine log schemes. Let $b\colon (T, \mathcal{M}_T) \to (S, \mathcal{M}_S)$ be another morphism of fine log schemes. Then,*

$$X^\dagger \times^{\mathrm{fine}}_{S^\dagger} T^\dagger \to X^\dagger \times_{S^\dagger} T^\dagger$$

is surjective and hence a homeomorphism on underlying topological spaces.

References

[1] Piotr Achinger, Katharina Hübner, Marcin Lara, and Jakob Stix. Logarithmic geometry beyond fs, 2024.

[2] Shinichi Mochizuki. Monomorphisms in categories of log schemes. *Kodai Math. J.*, 38(2):365–429, 2015.